Lecture 4: The Complexity Class **P**

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The Big–Oh Notation - Reminder

Definition

A function $f(n)$ is said to be $O(g(n))$ if and only if there are positive constants, c and n_0 such that $f(n) \leq c \cdot g(n)$ for all $n > n_0$.

Intuition: f grows not faster than g.

Exercise

1. $3n + 7$ is $O(n)$?
2. $n^2 + n$ is $O(n^2)$?
3. n^2 is $O(n^2 + n)$?
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5. $0.1n^2$ is $O(n)$?
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Question 1: compare growth rate of 2^n and 3^n; $2^n = o(3^n)$
Question 2: what about $\log_2 n$ and $\log_{999} n$?

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Question 1: compare growth rate of 2^n and 3^n; $2^n = o(3^n)$
Question 2: what about $\log_2 n$ and $\log_{999} n$?
$\log_2 n = \log_2 999 \cdot \log_{999} n \quad \Rightarrow \quad \log_2 n = \Theta(\log_{999} n)$

Time Complexity

Definition

For any function f, we say that the time complexity of a decidable language $\mathcal{L}$ is $O(f)$, or $\mathcal{L}$ is decidable in $O(f)$ time, if there exists a TM T which decides $\mathcal{L}$, and constants n_0 and c such that for all inputs x with $|x| > n_0$,

$$Time_T(x) \leq c \cdot f(|x|)$$

Complexity Classes

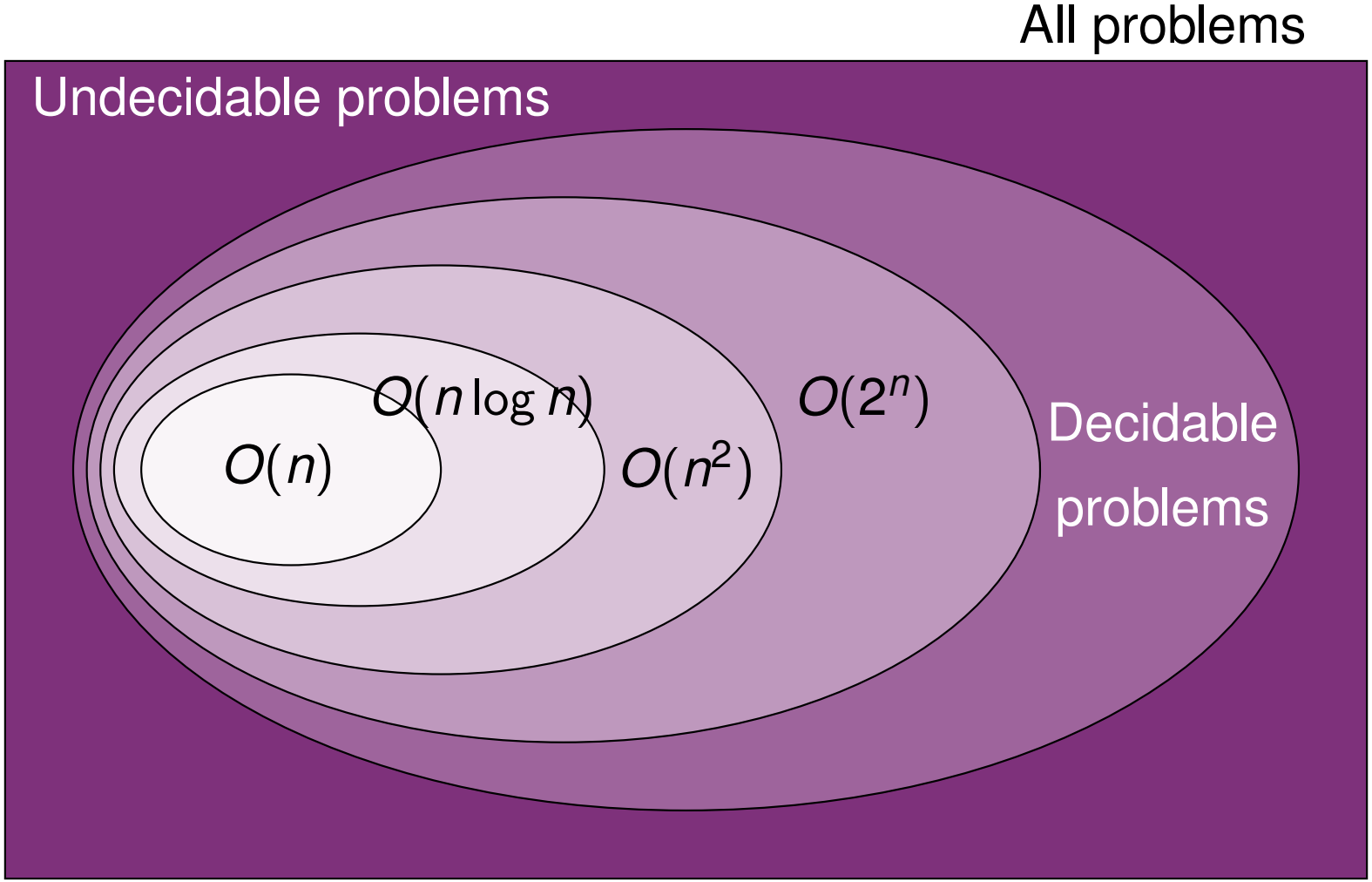
Now we are in a position to divide up the decidable languages into classes, according to their time complexity.

Definition

The time complexity class TIME[f] is defined to be the class of all problems (i.e. languages), for which there exists an algorithm with time complexity in $O(f)$.

- it is sometimes called DTIME[f] – for Deterministic Time

Examples


All problems
Undecidable problems
$O(n \log n)$
$O(2^n)$
Decidable problems
$O(n)$
$O(n^2)$

The Complexity Class **P**

Definition

$$\mathbf{P} = \bigcup_{k \geq 0} TIME[n^k]$$

Note: The complexity class **P** contains all classes $TIME[t(n)]$, where $t(n)$ is a polynomial of finite degree.

The Complexity Class **P**

Time Complexity	Size n					
	10	20	30	40	50	60
n	.00001 seconds	.00002 seconds	.00003 seconds	.00004 seconds	.00005 seconds	.00006 seconds
n^2	.0001 seconds	.0004 seconds	.0009 seconds	.0016 seconds	.0025 seconds	.0036 seconds
n^3	.001 seconds	.008 seconds	.027 seconds	.064 seconds	.125 seconds	.216 seconds
n^5	.1 seconds	3.2 seconds	24.3 seconds	1.7 minutes	5.2 minutes	13.0 minutes
2^n	.001 seconds	1.0 seconds	17.9 minutes	12.7 days	35.7 years	366 centuries
3^n	.059 seconds	58 minutes	6.5 years	3855 centuries	2×10^8 centuries	1.3×10^{13} centuries

$\mathbf{P}$ = doable?

The class **P** is a reasonable mathematical model of the class of problems which are tractable or solvable in practice.

However, the correspondence is not exact:

- When the degree of the polynomial is high then the time grows so fast that in practice the problem is not solvable
- The constants may also be very large, for example $2^{2^{2^{2^{2^{10}}}}}$

(These problems usually seem to arise only in artificially created problems; for naturally defined problems the polynomials and constants are not too large)

Different Models of Computation

Lemma

We can simulate t steps of k-tape TM with an equivalent one-tape TM in $O[t^2]$ steps

Lemma

We can simulate t steps of a two-way infinite k-tape machine with an equivalent k-tape TM in $O[t]$ steps

Hence the class **P** is the same for all of these models of computation (and many others).

Different Encodings

Lemma

For any number n, the length of the encoding of n in base b_1 and the length of the encoding of n in base b_2 are related by a constant factor (provided $b_1, b_2 \geq 2$)

- What if we encode n in base $b_0 = 1$?

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- What if we encode n in base $b_0 = 1$?

Lemma

For any graph G, the length of the encoding of G as an adjacency matrix, and the length of the encoding of G as a list of edges, are both related by a polynomial factor to the number of vertices.

- Which is this polynomial factor?

Hence the class **P** is the same for these encodings (and many others).

Proving a problem is in **P**

- The class **P** is said to be robust — it does not depend on the exact details of the computational model or encoding.

$\Rightarrow$ we do not need to specify all the details of the machine model or even the encoding (e.g. pseudo-code is sufficient).

- The most direct way to show that a problem is in **P** is to give a polynomial time algorithm which solves it.
- Even a naive polynomial-time algorithm often provides a good insight into how the problem can be solved efficiently.
- To find such an algorithm we generally need to identify an approach to the problem that is considerably better than brute-force search.

CNF

Some of the most important computational problems concern logical formulas

- A logical formula f is said to be in conjunctive normal form (CNF) if

$$f = C_1 \wedge \ldots \wedge C_m$$

  where each C_i is a clause, which is a disjunction ($\vee$, "OR") of literals:

$$C_i = (l_{i1} \vee \ldots \vee l_{ik})$$

  and a literal is either a variable or its negation, e.g. x_i or $\neg x_i$
- If f has at most k literals per clause then it is in k-CNF

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- If f has at most k literals per clause then it is in k-CNF
- For example:

$$(x_1 \vee x_2 \vee \neg x_5) \wedge (\neg x_2 \vee \neg x_4 \vee \neg x_5) \wedge (x_2 \vee x_3 \vee x_4)$$

  is in 3-CNF

Satisfiability

- The logical formula f is satisfiable if there exists an assignment of TRUE and FALSE to the variables of f which makes f TRUE .
- f is TRUE iff all clauses are TRUE
- a clause is TRUE iff at least one literal is TRUE
- For example:

$$(x_1 \vee x_2 \vee \neg x_5) \wedge (\neg x_2 \vee \neg x_4 \vee \neg x_5) \wedge (x_2 \vee x_3 \vee x_4)$$

is satisfied by this assignment (amongst others):

$$x_1 = \text{TRUE}\ , x_2 = \text{FALSE}\ , x_3 = \text{TRUE}\ , x_4 = \text{TRUE}\ , x_5 = \text{FALSE}\ .$$

The Satisfiability Problem

SATISFIABILITY
Instance: CNF formula f
Question: Is f satisfiable?

k-SATISFIABILITY
Instance: k-CNF formula f
Question: Is f satisfiable?

2-Satisfiability

Proposition

2-SATISFIABILITY *is in* **P**

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Proof:

0. Declare all clauses unsatisfied and literals unassigned.

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1. Select an arbitrary unassigned variable x and assign x the value TRUE and $\neg x$ the value FALSE

2-Satisfiability

Proposition

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Proof:

0. Declare all clauses unsatisfied and literals unassigned.
1. Select an arbitrary unassigned variable x and assign x the value TRUE and $\neg x$ the value FALSE
2. Select an unsatisfied clause $l_i \vee l_j$,
   1. If both literals are unassigned, ignore the clause.
   2. If at least one literal is assigned TRUE , declare the clause satisfied.
   3. if one literal is FALSE and the other is unassigned, set the other to TRUE and its negation to FALSE , and declare the clause satisfied.
   4. If both are FALSE , restart the algorithm setting x FALSE and $\neg x$ TRUE . If a conflict occurs again, declare unsatisfiable.

Proof continued

3. Repeat step 2, until either
   1. all clauses are satisfied, return satisfiable;

Proof continued

3. Repeat step 2, until either
   1. all clauses are satisfied, return satisfiable;
   2. or all clauses remaining not satisfied (yet) have all their variables unassigned. In this case return to step 1;

Exercise: Complete the proof by showing that this algorithm is correct and has polynomial running time.

Polynomial-time reducibility

Another way to show that a problem is in **P** is to use a reduction.

Informally, a problem P is reducible to a problem Q if we can somehow use methods that solve Q in order to solve P.

Definition

Language $\mathcal{L}_1$ is polynomially reducible to $\mathcal{L}_2$, denoted $\mathcal{L}_1 \leq \mathcal{L}_2$, if a polynomially-bounded and polynomial-time computable function f exists such that: $x \in \mathcal{L}_1 \Leftrightarrow f(x) \in \mathcal{L}_2$

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Recall: the (polynomial) reduction we saw from VERTEX COVER to SET COVER.

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This fundamental tool of complexity theory was introduced by Richard Karp in 1972.
(one of the inventors of the Edmonds-Karp algorithm for max flow).

Key property of polynomial-time reducibility

Lemma

$$\mathcal{L}_1 \leq \mathcal{L}_2 \textit{ and } \mathcal{L}_2 \in \mathbf{P} \Longrightarrow \mathcal{L}_1 \in \mathbf{P}$$

Main idea: The composition of polynomials is a polynomial.

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Main idea: The composition of polynomials is a polynomial.

Proof of the Lemma:

- Let A_2 be a polynomial-time algorithm that decides L_2
- Let f be a polynomial-time reduction algorithm from L_1 to L_2
- We construct a polynomial-time algorithm A_1 that decides L_1:

1. given input x of L_1, compute input $f(x)$ of L_2 in polynomial time (we know that $x \in L_1 \iff f(x) \in L_2$)
2. use algorithm A_2 to decide whether $f(x) \in L_2$
3. if $f(x) \in L_2$ then output YES; otherwise output NO

k-colorability

- Let $G = (V, E)$ be a graph, with vertices V and edges E.
- A function $f : V \to \{1, \ldots, n\}$ is a (proper) coloring if adjacent vertices are assigned different values (= colors).

k-COLORABILITY
Instance: A graph G
Question: Is there a coloring of G using at most k colors?

(When $k = 2$ this is equivalent to BIPARTITE)

2-COLORABILITY $\leq$ 2-SATISFIABILITY

We can reduce 2-COLORABILITY to 2-SATISFIABILITY:

- For each vertex v_i of the graph we create a variable x_i
- For each edge (v_i, v_j) we add two clauses $(x_i \vee x_j)$ and $(\neg x_i \vee \neg x_j)$.

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This translation of a 2-COLORABILITY problem to a 2-SATISFIABILITY problem is computable in polynomial time. Now we check it satisfies the reducibility condition:

$\Rightarrow$ If the graph is 2-colorable, use 2-coloring to assign truth values tovariables (one color is TRUE , the other FALSE).

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This translation of a 2-COLORABILITY problem to a 2-SATISFIABILITY problem is computable in polynomial time. Now we check it satisfies the reducibility condition:

$\Rightarrow$ If the graph is 2-colorable, use 2-coloring to assign truth values tovariables (one color is TRUE , the other FALSE).

$\Leftarrow$ If the formula is satisfiable, define 2-coloring by setting TRUE variables to color 1 and FALSE to color 2. If two adjacent vertices get the same color then one of the associated clauses is not satisfied (contradiction). Thus we have a proper 2-coloring.

An aside

- Some problems have been shown to be in **P** by non-constructive arguments (i.e., by proving a polynomial time algorithm must exist)
- e.g. The problem KNOTLESSNESS – can a given graph be embedded in 3-dimensional space without knots.
- In these cases we may not even know how to find an exponential algorithm!
- (See Bovet and Crescenzi for more discussion of this).